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Journal of Non-Crystalline Solids 263&264 (2000) 416–421

JOURNAL OF
NON-CRYSTALLINE SOLIDS

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Surface studies of semiconducting glass using ion beam methods

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Abstract

We have studied the near surface stoichiometry of a semiconducting vanadium phosphate glass for microchannel plate application using MeV ion beam techniques including Rutherford backscattering spectrometry (RBS), particle induced X-ray emission (PIXE) and elastic recoil detection (ERD) analysis. A conventional microchannel plate made of dielectric lead silicate glass with a surface conducting layer was also measured for comparison. The elemental profiles of the two samples showed that the semiconducting glass have a more uniform surface structure as compared to conventional microchannel plate glasses suggesting that the former is more stable. We suggest that our results indicate microchannel plates (MCPs) made of semiconducting glass should have long-term stability and therefore should have longer service life. © 2000 Published by Elsevier Science B.V. All rights reserved.

1. Introduction

Microchannel plates (MCPs) are arrays of channel electron multipliers which are the basic building blocks for a number of devices making use of the electron multiplication concept. They are used as detectors for ions and photons as well as image intensifiers for night vision applications [1–4]. Commercial MCPs are fabricated from insulating dielectric glasses, such as lead silicate glass. The glass materials are hydrogen-treated to produce an electronic conducting surface layer covering the channel walls.

The surface layers of the hydrogen-treated glass are in an activated state, and will gradually degrade during device operation [5]. Its ability to

emit and replenish electrons gradually decreases resulting in a limited service life for the MCP. Moreover, a large amount (a few at.%) of hydrogen is assumed to be present in these surface layers [5], either in the form of trapped hydrogen molecules or as the reaction products, water or other hydrogen groups. During operation, H₂O and H₂ enters the microchannels in the form of positively charged gas ions that may accelerate in reverse direction and damage the photocathode. The resistivity intrinsic to the matrix glass materials also limits the dynamic range of the devices. Investigators have suggested that for MCP applications it is desirable to replace the surface-conducting dielectric glass with a semiconducting or bulk-conducting (BC) glass [6,7].

Some oxide glasses containing transition metal ions, such as V⁵⁺/V⁴⁺ and Fe²⁺/Fe³⁺, have electronic conduction, the conduction mechanism of these ions in oxide glasses is assumed to be due to

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the hopping process as explained by the Mott theory [8]. The resistivity of these glasses can be changed in the range of 10^7 – 10^{12} Ω cm by selecting suitable chemical compositions. It was also found that certain BC glasses, such as vanadium phosphate glasses have electron multiplication capability (secondary electron emission coefficient >1) and may be used as the working layer of an electron multiplier tube [9].

In a BC glass, Joule heating is distributed over the bulk rather than over the surface layer of a few tens of nanometer. The temperature gradient should be more gradual in such materials. As a result differential thermal stress is less than that in the surface-conducting glass. Since the conductivity of BC glass is an electronic bulk property, we expect MCPs fabricated from this material to maintain their conductivities and have a longer service life.

Using BC glass material as matrix for MCPs, an inherently non-axial electric field inside the microchannel can be set up, stopping the positive ions drifting back toward the photocathode, eliminating the ion-feedback problem in MCPs [6,7]. Moreover, MCPs made from BC glass will have fewer H_2 or H_2O molecules and therefore the number of positive ions will be a minimum. The conductivities of BC glass also enable devices to work in a larger dynamic range. The steady and stiff boundary condition of the electric field inside the microchannels makes the BC glass MCP capable of detecting high frequency input signals [7]. The unique properties of BC glass, therefore make it a promising material for high yield, long lifetime MCPs.

A vanadium phosphate-based BC glass has been prepared, drawn into fibers and has been formed into microchannels [10]. In a previous paper we reported the electrical properties of the MCP material as well as the electron gain properties of MCP prototypes with various thickness and diameter fabricated from this vanadium phosphate glass [10]. In this report, we compare the near surface stoichiometry of a BC glass and a conventional MCP lead silicate glass using ion beam methods. The near surface region is of particular interest because the long-term stability of MCPs is assumed to be due to the change at the

working surface of the channel materials. Ion beam techniques have been utilized in the past for analyzing lead silicate glasses films [11,12] and proved to be useful for near surface compositional analysis.

2. Experimental procedures

The BC glass we used for this study is an iron–lead–vanadium phosphate glass, its near surface composition is shown in Table 1. The systems (P_2O_5 – V_2O_5) and (P_2O_5 – Fe_2O_3) have a wide glass-forming composition range [13,14]. PbO is added to these glasses to improve processing properties.

MCP prototypes were fabricated from the BC sample materials. The sample MCP disks were 25 mm diameter with hollow channel diameters of 15 μ m. Various disk thicknesses were made to give a range of channel length-to-diameter ratios ($40 \leq l/d \leq 100$). 576 microchannels were made on the center part of the disk. The axis of the channels was at an angle of 8° (bias angle) with respect to the normal direction of the two faces of the disks [10]. The resistivity of the sample was such that a disk current of several μ A could flow with 1000 V applied to the face electrodes. Preliminary electron gain measurements were carried out using a tantalum thermionic cathode placed at a distance of 4 cm from the MCP input face in a vacuum chamber at a pressure $\sim 10^{-6}$ Torr. Details of the measurements were presented in Ref. [10].

Table 1

Atomic composition (in at.%) of the BC glass and the conventional MCP glass in the near surface region (<1 μ m) as measured by RBS, PIXE and ERD^a

Elements	Conventional MCP	BC glass
Pb	3.7	6.0
P	–	18
V	–	3.9
Fe	–	4.5
H	6	0.5
Si	24.4	–
K	3.5	–
Na	2.4	–
O	60	67.1

^a The error in the composition is $\sim \pm 5\%$ of the measured at.%.

The near surface stoichiometry of the matrix of a conventional MCP was also measured in this study for comparison. The test MCP was a hydrogen-treated lead silicate glass 32.74 mm in diameter and was a market product of Galileo Electro-Optics Corporation [15].

Ion beam analysis was carried out using ion beams generated by a 2.5 MeV Van de Graaff electrostatic accelerator. Rutherford backscattering spectrometry (RBS) measurements were performed using a 1.95 MeV $^4\text{He}^+$ beam [16]. A Si surface barrier detector was located at a backscattering angle of 165° with respect to the ion beam to collect the scattered He ions from the sample. 1.95 MeV $^4\text{He}^+$ and H^+ beams were used for particle induced X-ray emission (PIXE) experiments. A Si(Li) detector positioned at 30° with respect to the ion beam was used to collect the X-rays emitted by the target materials. The higher velocity of the H^+ ions (with the same energy) enables the elemental analysis of the target materials to a depth of $>10\text{ }\mu\text{m}$. RBS and PIXE measurements were carried out simultaneously so that complementary information on depth profiling and elemental identification were obtained [16,17].

One of the key processes for the fabrication of MCPs using dielectric glass materials is hydrogen treatment to make a surface layer conducting. It is therefore important to know the hydrogen distribution in the near surface region of these materials. Hydrogen atomic distribution was obtained by elastic recoil detection (ERD) method using a 1.95 MeV $^4\text{He}^+$ [18]. This technique is also commonly known as the hydrogen forward scattering technique. In this type of measurement, the sample was tilted so that the incident and exit beams both made an angle of 15° with the sample surface. A Si surface barrier detector with $\sim 10\text{ }\mu\text{m}$ thick Mylar absorber positioned at 30° with respect to the ion beam in the forward direction was used to detect the forward recoiled hydrogen atoms in the sample with the forward scattered He ion absorbed by the Mylar layer [18]. Absolute quantities of hydrogen in the samples can be measured by the ERD technique [18]. In our experiment, a Kapton foil was also used as standard to improve the accuracy of the measurements.

3. Results

In our previous report, the conductivity (σ)-temperature (T) relationship of the BC glass showed a linear $\log [T \cdot \sigma]$ versus $1/T$ dependence with a slope corresponding to an activation energy of 0.51 eV [10]. This energy value falls within the range characteristic of the group of materials cited by Mott and Davis [8]. This temperature dependence conductivity follows Mott conduction by polaron interactions of transition metal ions in glass under electric field excitation [8].

Fig. 1 shows the electron gain result for a typical MCP prototype fabricated from the BC glass with a $l/d = 45$. The total input beam current was 120 nA; of which, 0.12 nA was intercepted in the channel cluster. We noticed that the electron gain increased exponentially with applied voltage, comparable to a commercial MCP with similar geometry [19]. Between 400 and 1800 V, the gain of the MCP increases by a factor of several thousand. A linear region in the exponential gain curve was observed between bias voltage of ~ 500 –1000 V (indicated by the two arrows on Fig. 1), which is the normal operating range for MCPs. The rather constant gain at voltages <400 V was caused by ammeter leakage current, which masked the real, but smaller gain in this region.

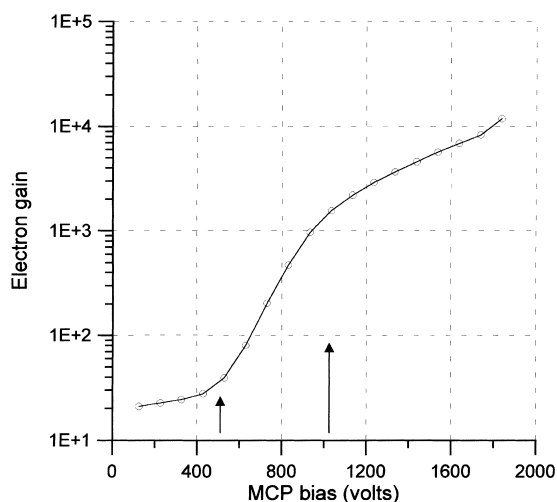


Fig. 1. The electron gain of a typical MCP fabricated from BC glass with $l/d = 45$.

Fig. 2 shows the RBS spectra from the BC glass and the hydrogen-treated lead silicate glass from a conventional MCP. The symbol spectra are the raw RBS data while the lines (solid and dashed) are the computer simulation [20] according to the elemental composition listed in Table 1. The Al signals in the two spectra are from a ~ 10 nm Al coating on the sample surface which prevents sample charging. The analyzable depth in the RBS experiment is ~ 1.5 μm .

Since there is an ambiguity in element mass and depth distribution in RBS we also accumulated the PIXE signals simultaneously during the RBS experiments. Fig. 3 shows the PIXE results from the two types of glasses. It should also be noted that the sensitivity of RBS for elements with $Z < 20$ is relatively poor due to interference by signals from larger Z elements and because the scattering cross section for RBS is proportional to Z^2 . The sensitivity of the PIXE method is inherently greater and can detect smaller Z elements at atomic fraction of $\sim 10^{-6}$. In Fig. 3 it can be noticed that no trace impurities ($>10^{-6}$) other than the main constituents listed in Table 1 were detected in either sample. Additional PIXE experiments using 1.95 MeV H^+ beam gave similar X-ray spectra indicating that the elemental distributions in both types of glass were uniform to ~ 10 μm below the surface.

The hydrogen distributions of the two samples measured by elastic recoil detection (ERD) are

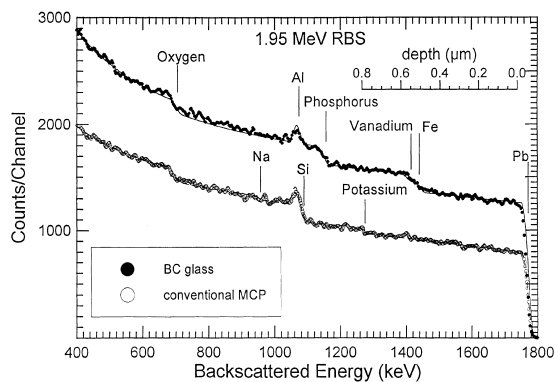


Fig. 2. RBS spectra from the BC glass and the hydrogen-treated lead silicate glass from a conventional MCP. Calculated spectra according to the elemental composition listed in Table 1 are also shown (lines) for comparison.

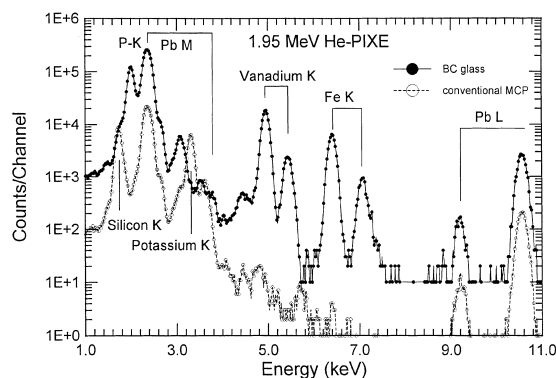


Fig. 3. PIXE results obtained simultaneously with RBS from the two types of glasses. The spectrum from the BC glass is shifted by a decade in the figure.

displayed in Fig. 4. The solid square and open circle data correspond to the ERD spectra from the conventional MCP sample and BC sample, respectively. ERD spectrum from a crystalline Si sample is also included (dashed line) to show the system noise level. The peak around channel 390 (~ 0.7 MeV) in the spectrum from the Si sample is produced by a hydrocarbon layer adsorbed on the surface of the sample. The signal drop-off at channel 100 arises from the electronics and was

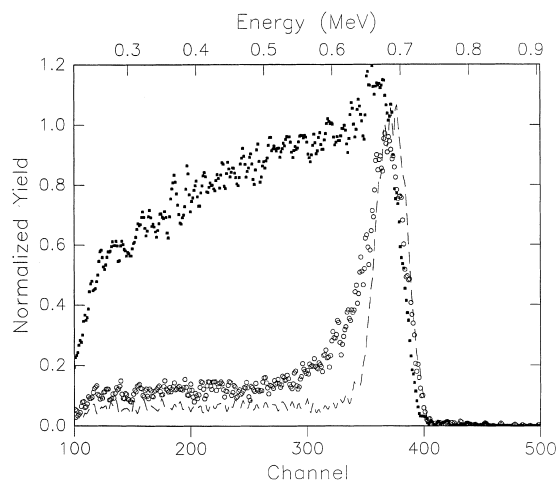


Fig. 4. Hydrogen signals obtained by ERD from the conventional MCP glass and BC glass samples. ERD result from a bare crystalline Si is also shown to establish the base line for the signals.

not the result of reduced hydrogen [18]. The analyzable depth in this experiment is $\sim 0.4 \mu\text{m}$.

4. Discussion

ERD from the BC sample showed that the hydrogen in the sample were only slightly higher than the noise level given in the ERD spectrum of the crystalline Si. Computer simulation of the spectrum revealed that $0.5 (\pm 0.2)$ at.% of hydrogen were present in the BC glass uniformly throughout the analyzed depth. The conventional MCP sample, on the other hand had a hydrogen content of 6.5 ± 0.5 at.% as observed from the ERD measurement. The hydrogen content in the conventional MCP sample also decreased as a function of depth from 6.5 at.% at the surface to 5.5 at.% at $\sim 0.4 \mu\text{m}$ below the surface.

The nominal atomic composition of both types of glass as measured by RBS, PIXE and ERD are summarized in Table 1 and the detail elemental distributions are plotted in Fig. 5. The various elements in the BC sample were distributed uniformly in the near surface region. The elemental distribution of the conventional MCP sample, however, is slightly inhomogeneous. A depletion of Pb was observed at the surface while hydrogen decreased as a function of depth. It should also be mentioned that for the BC glass samples, the surface contacting the leachable core glass should be identical to the actual channel surface.¹ For the conventional MCP sample, we assume that both the channel wall and the outer surface are similar.

The near surface stoichiometry of lead silicate glass has been studied by Then and Pantano [5]. Using low energy ion scattering (ISS) and secondary ion mass spectrometry (SIMS), Then and Pantano found that lead silicate glass after hy-

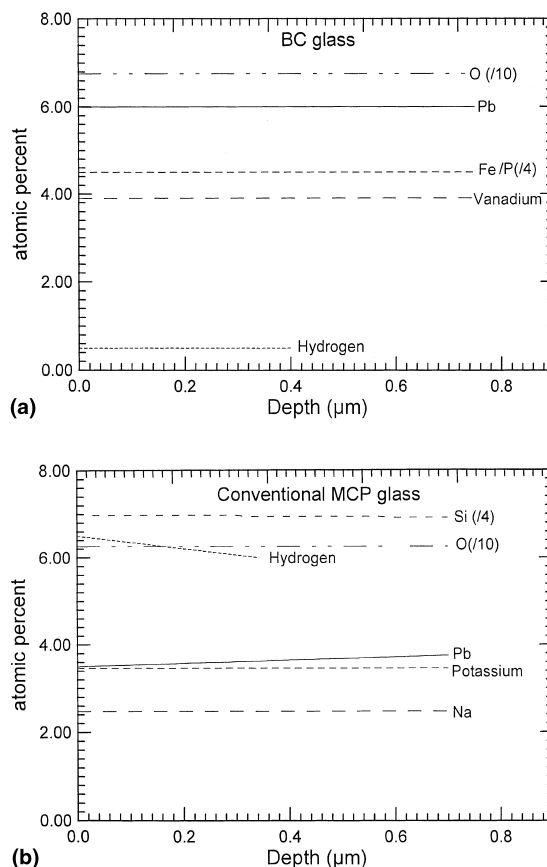


Fig. 5. Atomic distribution of the two types of glass obtained by RBS, PIXE and ERD measurements.

drogen reduction formed an ~ 30 nm silica-rich surface layer, and at depths >30 nm a layer depleted in Pb atoms. About 2 at.% of hydrogen were measured by Then and Pantano extending to $\sim 0.5 \mu\text{m}$ below the surface. From our investigation on the conventional MCP glass sample a rather uniform distribution of the various constituent elements was observed with only a slight Pb depletion on the sample surface. A silica-rich surface layer with a depth of <20 nm, however, cannot be detected by RBS [18].

We should also point out the samples studied by Then and Pantano were prepared in the laboratory while our sample was a commercial product. Moreover, the techniques we used for the present work differ from those employed by Then and Pantano and therefore the two sets of results

¹ We also measured samples from both: (1) the surface contacting with the leachable core glass layer (by leaching away the core glass layer, the sample resembles the working surface of the microchannel wall), and (2) the 'fire-polished' free surface of the BC glass. These two BC glass samples had identical elemental profiles indicating that the BC glass was stable during processing and was compositionally homogeneous throughout the bulk.

cannot be directly compared. While SIMS is a technique for measuring trace impurities, the absolute quantification of the technique is rather poor [17]. We assume the much higher hydrogen content we measured by the quantitative ERD technique is a more reliable result.

5. Conclusion

Our ion beam measurements on BC and conventional MCP glass samples show that the composition of the vanadium phosphate BC glass is uniform to depths from the surface to $\sim 10\ \mu\text{m}$. Elemental distributions in the near surface region of the lead silicate glass in the conventional MCP were inhomogeneous. In addition, about 6 at.% of H was measured in the conventional MCP glass while only ~ 0.5 at.% H was present in BC glass. We conclude from these results that the BC glass has a more stable surface structure than conventional MCP glass. Therefore we assume that MCPs made of BC glass should be more stable and hence should have longer service life.

Acknowledgements

The authors thank Dr T. Huen for setting up the electrical measurement apparatus and assistance in performing the electron gain measurements. The ion beam works performed at LBNL were supported by the Director, Office of Science, Office of Basic Energy Sciences, Materials Science Division of the US Department of Energy.

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